Why exact connector contact cannot use a clean equal-arm cross in the current Wavevis overhang

Wavevis inverse-sheet constraint note

June 4, 2026

Main result. For a four-arm cell with a fixed center C and four fixed contacted connector points, a clean cross with exact contact and one common half-arm length L exists only in a special symmetric geometry. The current Wavevis overhang is not that geometry: one witness cell requires arm lengths 0.327, 0.411, 1.243, and 1.576 at the same time. A second witness fails the through-center midpoint test by about 0.249. **Therefore:** with exact connector contact and no zig-zag legs as hard constraints, the common all-four arm length must be relaxed. If perpendicularity is also hard, that must be relaxed too.

Claim. For a four-arm inverse-sheet cell with a fixed center, four prescribed contacted connector points can be reached by a clean equal-length cross only if those four points satisfy exact midpoint, equal-radius, and, when requested, perpendicularity conditions. The current Wavevis overhang connector geometry violates those necessary conditions. Therefore no solver can satisfy exact connector contact, non-zig-zag arms, all-four equal arm length, and perpendicular crossed pairs at the same time for the current overhang.

Why this is a kinematic proof, not a solver preference. Exact contact fixes the endpoint positions. A clean non-zig-zag cross fixes the endpoint symmetries around the cell center. Equal arm length fixes all four endpoint distances to one value L. These are coordinate-invariant facts; rotating the local axes, changing the numerical optimizer, or smoothing the drawing cannot make unequal distances equal.

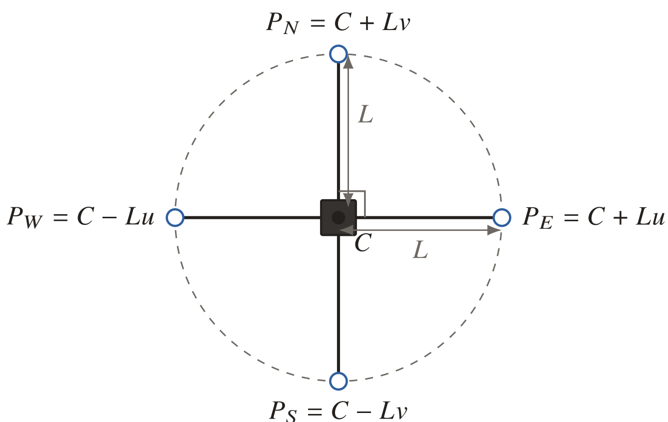
Scope. This is not a claim that a flat square grid can never satisfy these constraints. A perfectly symmetric flat cell can. The result here is specific to the prescribed connector positions in the current curved Wavevis overhang. Once those contacted connector points are fixed, the requested clean equal-length cross is algebraically over-constrained.

Question	Short answer	Reason
Can a different solver fix it?	No, not without changing a constraint.	The contradiction is L needing to equal four different measured distances for one cell.
Can dropping perpendicularity fix it?	No.	Equal arm length alone still requires all four contacted points to be the same distance from C.
What relaxation keeps the mechanism clean?	Pairwise equal bars.	East-west may use length L_x; north-south may use length L_y; the angle between the bars is free.

0. Visual map of the proof

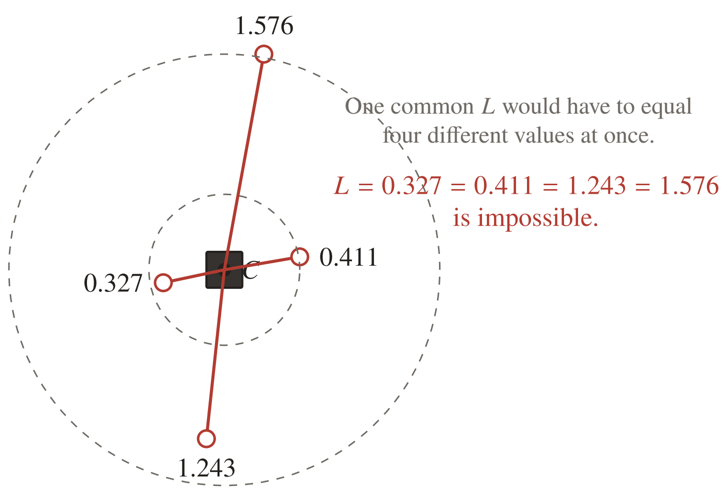
The whole question reduces to whether the four contacted connector points assigned to one cell have the special symmetry that an equal-length through-cross demands. The diagrams below show the tests before the algebra.

Requested clean cell	Algebraic consequence	What the current overhang does
Four arms have one common length L.	All four contacted connector points must be the same distance from C.	A current witness cell needs four different distances: 0.327, 0.411, 1.243, and 1.576.
Opposite legs are not zig-zagged.	Each opposite connector pair must have midpoint exactly C.	A current witness midpoint misses the required center by about 0.249.
Two pairs stay perpendicular.	The two through-cell directions must have dot product zero.	This is an additional restriction on top of the already-failed equal-radius and midpoint tests.



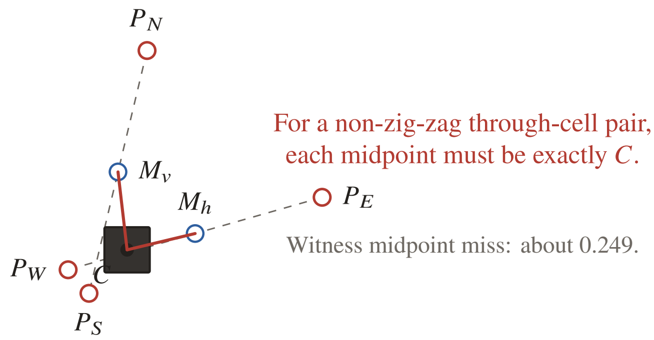
Equal-length perpendicular cross: all four connectors lie on one circle/sphere centered at C, and opposite pairs average to C.

Figure 1: The special geometry that would make all requested hard constraints true. It is possible only when the contacted connector points already have this symmetry.



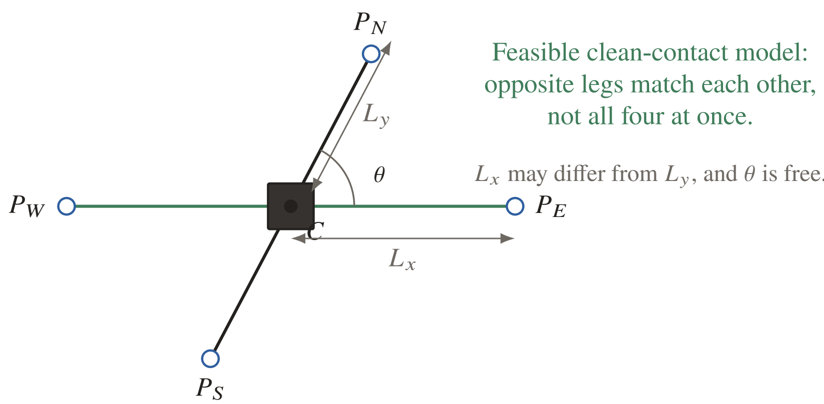
Current overhang witness cell n-10-26: exact contact forces unequal center-to-connector distances.

Figure 2: The equal-radius test fails in the current Wavevis overhang. This alone rules out both the perpendicular equal-length cross and the non-perpendicular equal-length cross.



If $M_h \neq C$ or $M_v \neq C$, then the contacted endpoints cannot be opposite straight arms through the cell center.

Figure 3: The midpoint test is independent of the length test. It checks whether opposite contacted connectors can form a straight bar through the cell center.



This removes the incompatible all-four common-length and right-angle demands while preserving clean pairwise bars.

Figure 4: The relaxed model that can remain visually clean: each opposite pair is collinear and equal within the pair, but the two pair lengths and included angle are free.

1. Definitions

Consider one cell with center point $C \in \mathbb{R}^3$. Let the four contacted connector points assigned to that cell be

$$P_E, \quad P_W, \quad P_N, \quad P_S.$$

"Contact" means the visible arm endpoint equals the prescribed connector point. Therefore the four arm vectors are

$$e = P_E - C, \quad w = P_W - C, \quad n = P_N - C, \quad s = P_S - C.$$

The phrase "avoid weird zig-zag legs" is treated as a hard visual/kinematic constraint:

opposite arms must be straight through the cell center.

That means the east and west arms lie on one line through C, and the north and south arms lie on another line through C.

2. Hard equal-length cross with perpendicular pairs

This is the strongest requested model. It requires unit directions u and v, one common half-arm length $L > 0$, and perpendicular arm-pairs:

$$P_E = C + Lu, \quad P_W = C - Lu, \quad P_N = C + Lv, \quad P_S = C - Lv, \quad u \cdot v = 0.$$

Necessary conditions. If these equations hold, then:

$$\frac{P_E + P_W}{2} = C, \qquad \frac{P_N + P_S}{2} = C \tag{1}$$

$$\|P_E - C\| = \|P_W - C\| = \|P_N - C\| = \|P_S - C\| = L \tag{2}$$

$$(P_E - C) \cdot (P_N - C) = 0. \tag{3}$$

Why. Equation (1) follows because $C + Lu$ and $C - Lu$ average to C, and similarly for v. Equation (2) follows because every vector is either Lu, $-Lu$, Lv, or $-Lv$, all with norm L. Equation (3) is perpendicularity.

Sufficient conditions. Conversely, if (1), (2), and (3) hold, then define

$$u = \frac{P_E - C}{L}, \qquad v = \frac{P_N - C}{L}.$$

The conditions reconstruct the exact perpendicular equal-length cross. So (1)–(3) are not just nice-to-have checks; they are the exact algebraic test.

Conclusion. If any one of (1), (2), or (3) fails for the contacted connector points of a cell, then exact contact and this clean equal-length perpendicular cross are kinematically impossible for that cell. No rotation of the cross, choice of local axes, or numerical solve can remove that contradiction while the connector points and cell center remain fixed.

3. Equal-length cross without requiring perpendicular pairs

Now remove the perpendicular condition. This is weaker:

$$P_E = C + Lu, \quad P_W = C - Lu, \quad P_N = C + Lv, \quad P_S = C - Lv$$

where u and v do not have to be perpendicular.

The exact necessary and sufficient conditions become only:

$$\frac{P_E + P_W}{2} = C, \qquad \frac{P_N + P_S}{2} = C \tag{4}$$

$$\|P_E - C\| = \|P_W - C\| = \|P_N - C\| = \|P_S - C\| = L. \tag{5}$$

Perpendicularity is gone, but the common radius L remains. Therefore all four contacted connector points must still lie on one sphere centered at C, and opposite pairs must still be symmetric through C.

Conclusion. Dropping the right angle does not solve the main problem. If the four contacted connector distances from C are unequal, then no equal-length through-cross can contact them, even if the two arm-pairs are allowed to meet at any angle. The all-four equal-length condition alone forces all four contacted points onto one sphere centered at C.

4. Numerical witness from the current Wavevis overhang

For the current Wavevis inverse-sheet overhang, the contacted connector geometry was evaluated from the local model on June 4, 2026. One cell gives the following distances from its center to its four contacted connector points:

Cell	Connector edge	Distance from cell center
n-10-26	e-h-10-25	0.411
n-10-26	e-h-10-26	0.327
n-10-26	e-v-9-26	1.576
n-10-26	e-v-10-26	1.243

The spread between the shortest and longest required contacted arm lengths is

$$1.576 - 0.327 = 1.249.$$

If all four arms had one common length L, then the same cell would require

$$L = 0.411, \quad L = 0.327, \quad L = 1.576, \quad L = 1.243$$

simultaneously. That is a contradiction. Therefore, for this cell, exact connector contact and equal length of all four arms cannot both be true.

This contradiction already proves both impossibility statements:

- The perpendicular equal-length cross is impossible, because it requires equal distances.
- The non-perpendicular equal-length cross is also impossible, because it still requires equal distances.

A second independent check shows the through-center symmetry condition is also violated in the current contacted layout. For cell n-8-21, the midpoint disagreement between the opposite-pair connector midpoints and the cell center is approximately

$$0.249.$$

For any exact through-line pair, that value must be exactly zero:

$$\frac{P_E + P_W}{2} = C, \qquad \frac{P_N + P_S}{2} = C.$$

So the current connector layout also fails the required midpoint condition for a clean through-cell model.

5. What relaxed model is actually feasible

A clean contacted cell can exist if the two opposite arm-pairs are treated as two separate bars:

$$P_E = C + L_x u, \quad P_W = C - L_x u$$

$$P_N = C + L_y v, \quad P_S = C - L_y v$$

where:

$$L_x \neq L_y \quad \text{is allowed}$$

and

$$u \cdot v \neq 0 \quad \text{is allowed.}$$

This model still preserves the important clean visual constraint:

- east and west are collinear and equal to each other;
- north and south are collinear and equal to each other;
- the cell looks like two straight bars crossing at the body, not like four random zig-zag spokes.

The exact algebraic test for this relaxed model is:

$$\frac{P_E + P_W}{2} = C, \qquad \|P_E - C\| = \|P_W - C\| = L_x \tag{6}$$

$$\frac{P_N + P_S}{2} = C, \qquad \|P_N - C\| = \|P_S - C\| = L_y. \tag{7}$$

No condition requires $L_x = L_y$, and no condition requires $u \cdot v = 0$.

Interpretation. This is the first clean-contact model with enough freedom to follow a curved overhang without forcing visible zig-zag arms. It preserves pairwise straightness, but gives up the single common length and the fixed right angle. In practice, exact contact must be solved with this pairwise model: the east-west contact pair and the north-south contact pair each need their own length, and the included angle between the pairs must be free.

If the pairwise midpoint equations in (6) and (7) are also impossible for a chosen fixed center, then the remaining choices are mechanical, not cosmetic: move the center, move the contacted connector assignment, allow a contact residual/gap, or allow a zig-zag/non-collinear leg. A smooth renderer cannot make an incompatible set of endpoint constraints true.

6. Engineering conclusion

For the current Wavevis overhang, exact connector contact and a clean visible mechanism can be made compatible only by relaxing the constraints in the correct order:

1. Keep exact connector contact if contact is physically required.
2. Keep opposite-pair collinearity if avoiding zig-zag legs is visually and mechanically required.
3. Allow the east-west pair length L_x to differ from the north-south pair length L_y.
4. Allow the two pair directions to meet at a non-right angle.

Constraint that must be removed: one common half-arm length for all four arms. If perpendicularity is also kept hard, it must be removed as well. Otherwise the contacted connector points must satisfy special equal-radius, symmetric, perpendicular conditions that the current wave geometry does not satisfy.

Therefore: with exact contact and no zig-zag legs as hard constraints, the feasible mechanism is a two-bar crossed cell with pairwise equal opposite legs, separate pair lengths, and a free included angle.